Data Reporting Automation

1. Project

Data Cleaning & Reporting Automation Using Python

2. Objective

The main goal of this Data Cleaning & Reporting Automation project is to automate the process of cleaning employee data and generating reports using Python. This Data Cleaning & Reporting Automation project improves data quality by removing records handling missing values standardizing inconsistent data and creating visual reports that help organizations analyze employee information efficiently.

3. Dataset Description

The dataset for this Data Cleaning & Reporting Automation project contains employee information with the following fields:

- Employee_ID – ID of each employee
- Name – Employee name
- Department – Department in which the employee works
- City – Employee location
- Age – Employee age
- Salary – Employee salary
- Experience – Years of work experience
- Performance_Rating – Employee performance score

This dataset was created in Microsoft Excel. Imported into Python for data cleaning, preprocessing and report generation for the Data Cleaning & Reporting Automation project.

4. Data Preprocessing

For the Data Cleaning & Reporting Automation project the following preprocessing steps were performed:

- Imported the dataset into Python using Pandas.
- Removed duplicate records from the dataset for the Data Cleaning & Reporting Automation project.

- Filled missing values using the column mean for the Data Cleaning & Reporting Automation project.
- Filled missing text values with "Unknown" for the Data Cleaning & Reporting Automation project.
- Standardized text by removing spaces and converting text to title case for the Data Cleaning & Reporting Automation project.
- Saved the cleaned dataset for analysis of the Data Cleaning & Reporting Automation project.

5. Tools & Technologies Used

- Python
- Pandas
- Matplotlib
- Microsoft Excel
- Google Colab / Visual Studio Code

6. Data Cleaning Techniques Used

The following techniques were applied to improve data quality for the Data Cleaning & Reporting Automation project:

- Duplicate Record Removal
- Missing Value Handling
- Text Standardization
- Data Validation
- Summary Statistics Generation
- Automated Report Generation

7. Reports and Visualizations Generated

The Data Cleaning & Reporting Automation project automatically generates the following reports and charts:

- Employees by Department
- Average Salary by Department
- Employee Distribution by City
- Age Distribution
- Salary Distribution
- Experience vs Salary

- Performance Rating Distribution

These visualizations provide insights into employee demographics, salary trends, departmental distribution and performance for the Data Cleaning & Reporting Automation project.

8. Business Insights

The analysis of the Data Cleaning & Reporting Automation project provides the following insights:

- Employee distribution across departments can be easily identified.
- Average salary varies across departments.
- Employee locations are distributed across cities.
- Salary generally increases with experience.
- Performance ratings help identify employee performance levels.
- Automated reports reduce effort and improve reporting accuracy for the Data Cleaning & Reporting Automation project.

9.Conclusion:

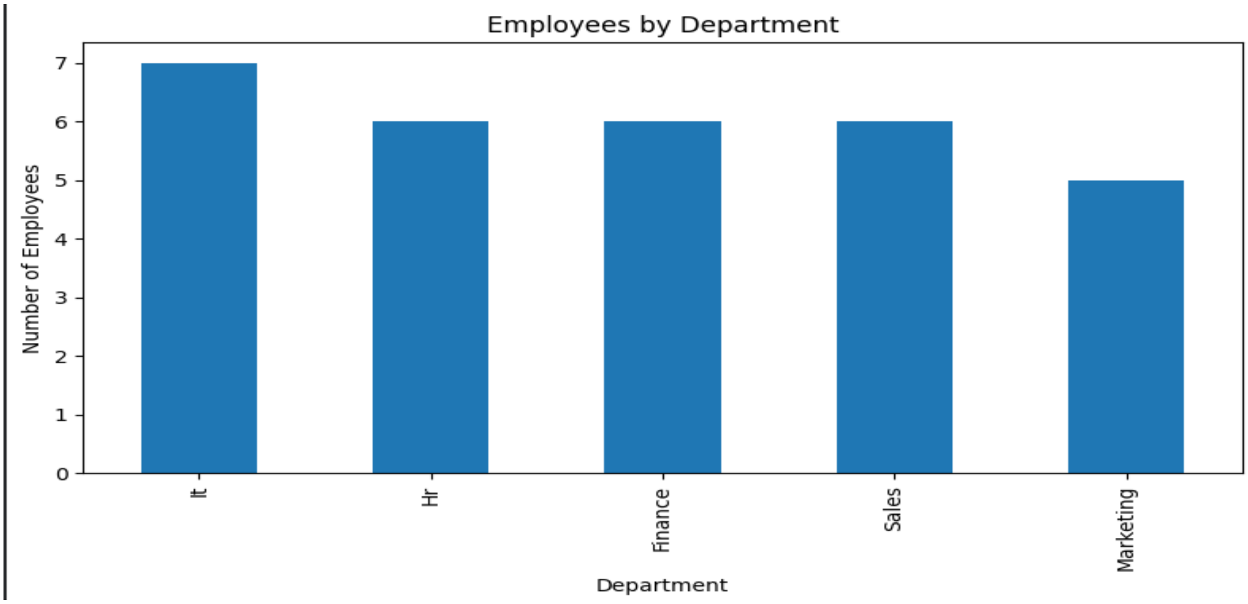
This Data Cleaning & Reporting Automation project demonstrates how Python can automate data cleaning and reporting processes. By removing records handling missing values standardizing data and generating automated reports with visualizations the Data Cleaning & Reporting Automation project improves data quality and reporting efficiency. The generated insights help organizations make decisions regarding workforce management and business planning, for the Data Cleaning & Reporting Automation project.

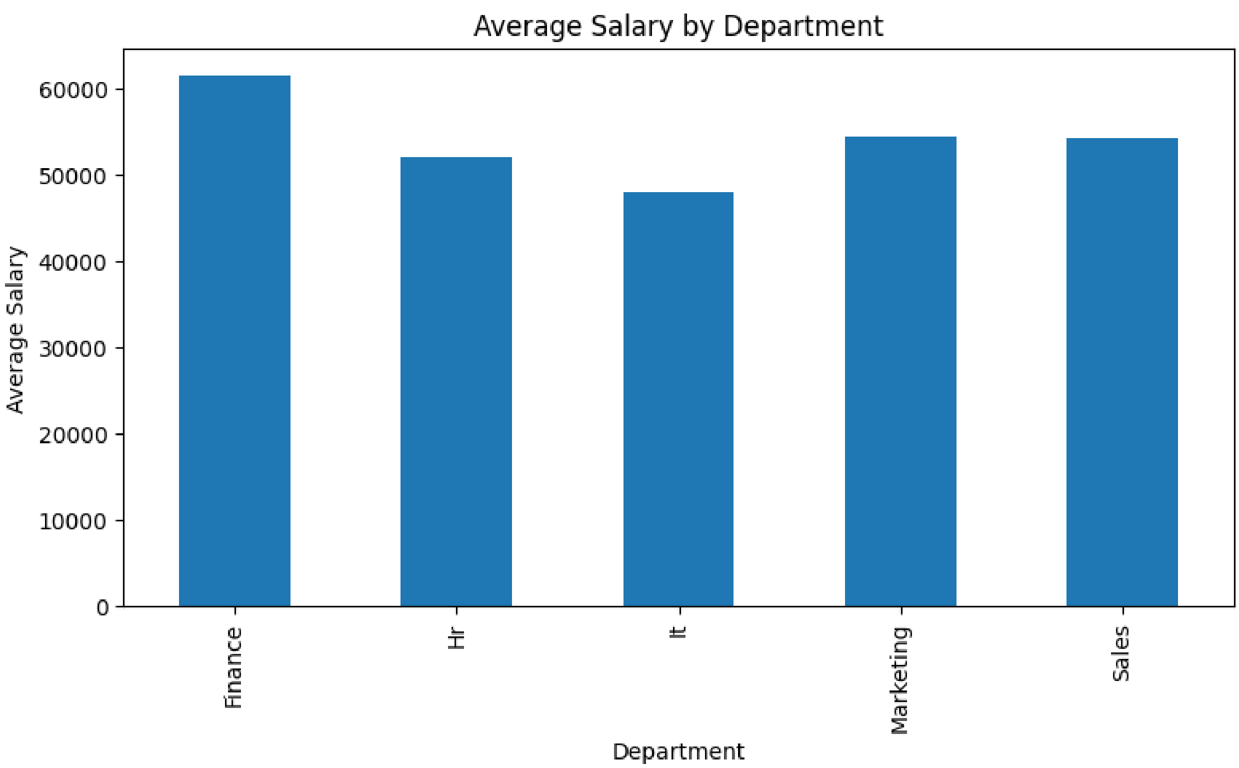
Outputs:

```
========== Original Dataset ==========
   Employee_ID   Name Department       City  Age  Salary  Experience  \
0          101  Rahul         IT  Hyderabad   25   45000           2
1          102  Priya         HR  Bangalore   29   52000           4
2          103  Arjun    Finance    Chennai   31   58000           6
3          104  Sneha         IT  Hyderabad   27   47000           3
4          105  Kiran      Sales     Mumbai   30   55000           5

   Performance_Rating
0                   4
1                   5
2                   3
3                   4
4                   5

Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Employee_ID         30 non-null     int64
 1   Name                30 non-null     object
 2   Department          30 non-null     object
 3   City                30 non-null     object
 4   Age                 30 non-null     int64
 5   Salary              30 non-null     int64
 6   Experience          30 non-null     int64
 7   Performance_Rating  30 non-null     int64
dtypes: int64(5), object(3)
memory usage: 2.0+ KB
None

Duplicate Rows: 0

Data Cleaning Completed Successfully!
```

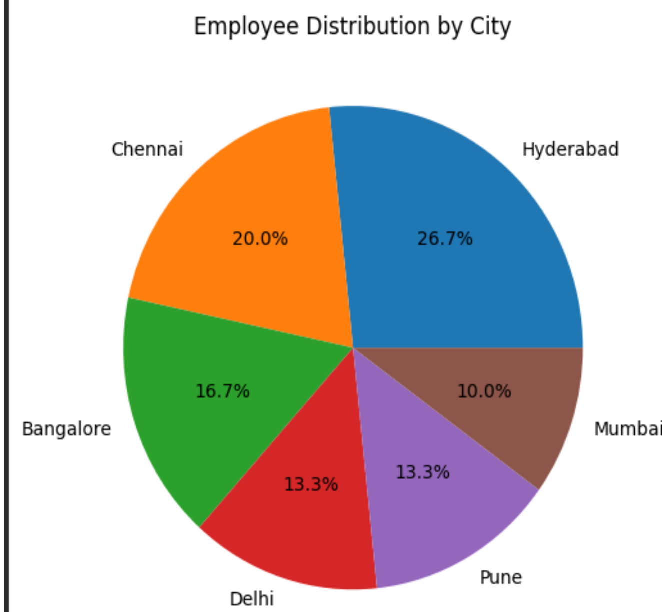

```
Duplicate Rows: 0

Data Cleaning Completed Successfully!
```

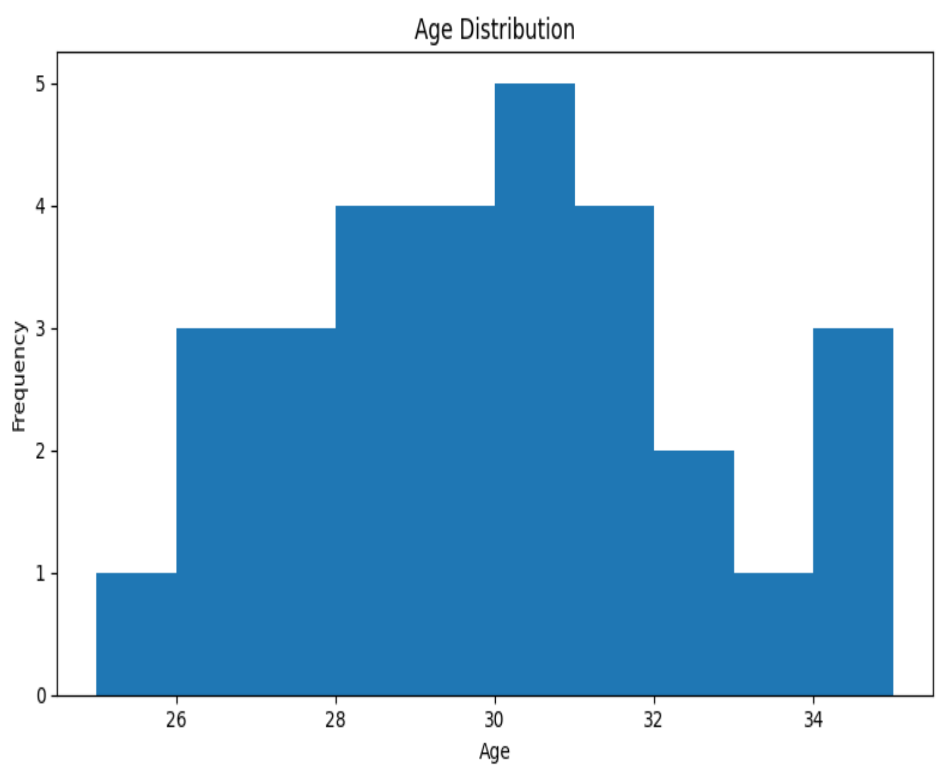

```
========== Summary ==========
        Employee_ID   Name Department       City        Age        Salary  \
count     30.000000     30         30         30  30.000000     30.000000
unique          NaN     30          5          6        NaN           NaN
top             NaN  Rahul         It  Hyderabad        NaN           NaN
freq            NaN      1          7          8        NaN           NaN
mean     115.500000    NaN        NaN        NaN  29.533333  53833.333333
std        8.803408    NaN        NaN        NaN   2.569494   5086.041299
min      101.000000    NaN        NaN        NaN  25.000000  45000.000000
25%      108.250000    NaN        NaN        NaN  28.000000  50250.000000
50%      115.500000    NaN        NaN        NaN  29.500000  53500.000000
75%      122.750000    NaN        NaN        NaN  31.000000  56750.000000
max      130.000000    NaN        NaN        NaN  35.000000  65000.000000

        Experience  Performance_Rating
count    30.000000           30.000000
unique         NaN                 NaN
top            NaN                 NaN
freq           NaN                 NaN
mean      4.900000            4.200000
std       1.768498            0.664364
min       2.000000            3.000000
25%       4.000000            4.000000
50%       5.000000            4.000000
75%       6.000000            5.000000
max       9.000000            5.000000
```


Employees by Department
Number of Employees
0
1
2
3
4
5
6
7
It
Hr
Finance
Sales
Marketing
Department

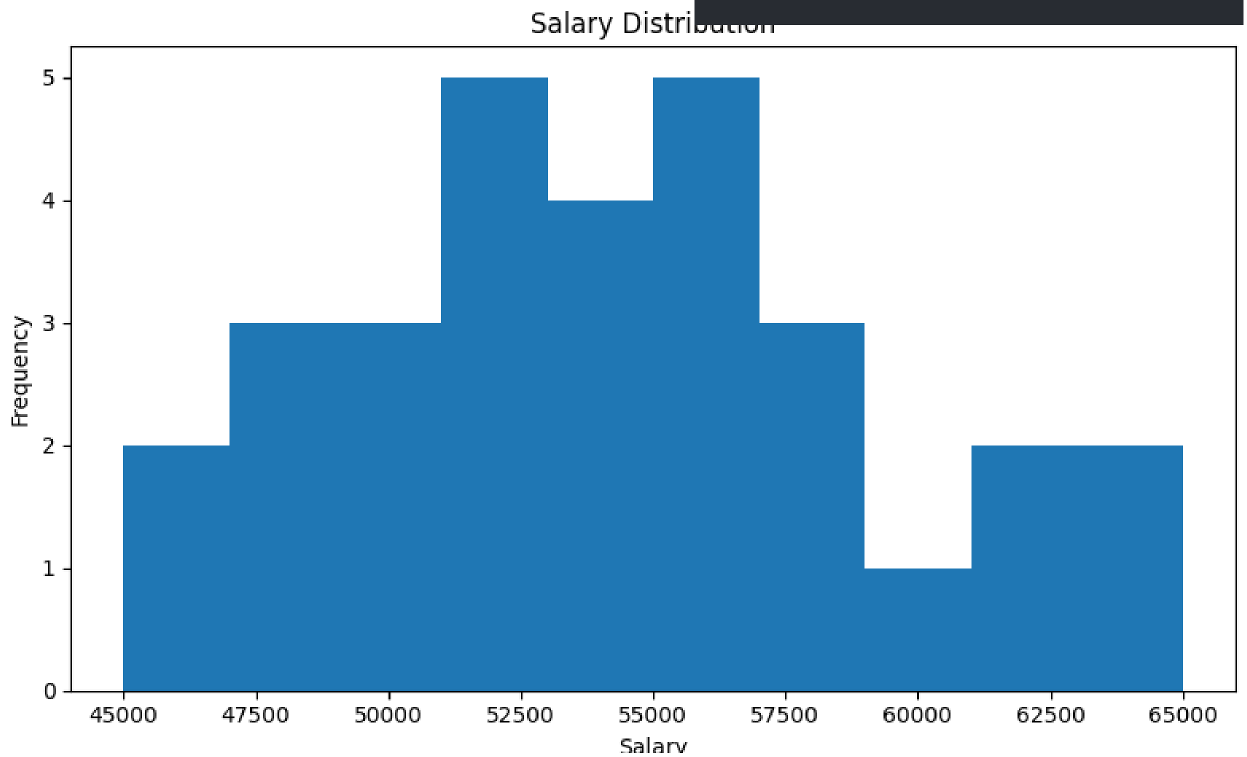

Average Salary by Department
Average Salary
0
10000
20000
30000
40000
50000
60000
Finance
Hr
It
Marketing
Sales
Department

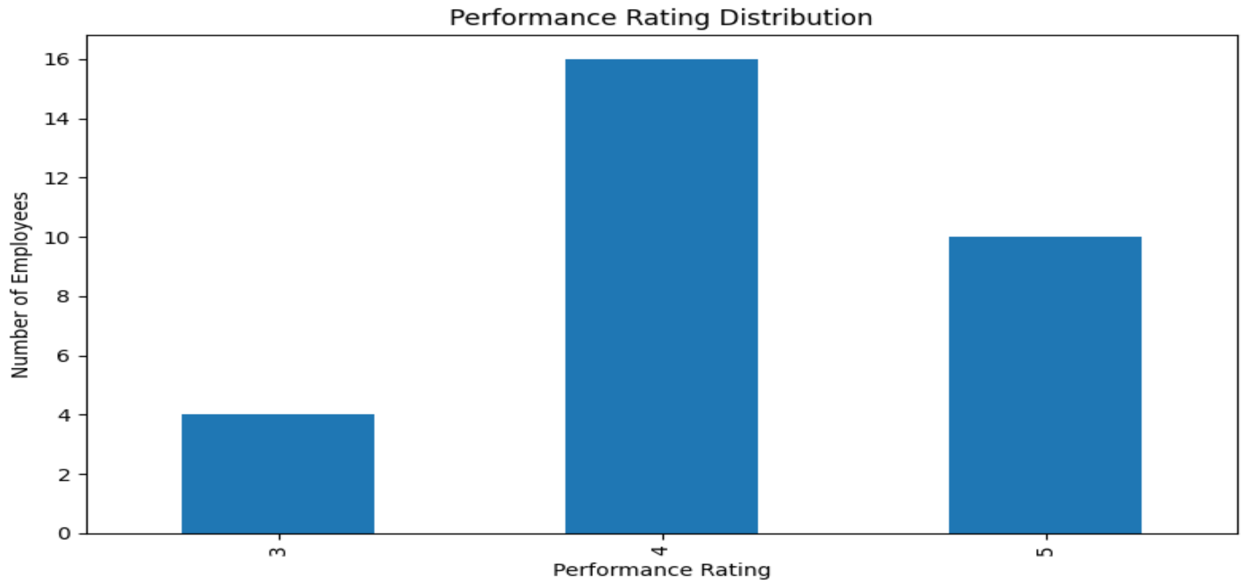
Department


Employee Distribution by City
Hyderabad
26.7%
Chennai
20.0%
Bangalore
16.7%
Delhi
13.3%
Pune
13.3%
Mumbai
10.0%


Age Distribution
Frequency
0
1
2
3
4
5
26
28
30
32
34
Age

colab.research.google.com – To exit full screen






```
====================================
Automation Completed Successfully!
====================================

Generated Files
1. Cleaned_Employee_Data.xlsx
2. Summary_Report.xlsx
3. Employees_By_Department.png
4. Average_Salary_Department.png
5. Employees_By_City.png
6. Age_Distribution.png
7. Salary_Distribution.png
8. Experience_vs_Salary.png
9. Performance_Rating.png
```